

Nidhi Agarwal

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SUMMARY

- Technical leadership experience in enterprise strategy, large-scale modernization, developer productivity, AI adoption, and distributed platform transformation.
- Educational background: Bachelor of Engineering in Computer Science; MBA, University of Connecticut.
- Currently leading enterprise modernization, capacity & performance engineering, engineering productivity, cloud-native transformation, and platform strategy at Walmart.
- Track record establishing engineering standards and enterprise platforms across Walmart, Microsoft, RBC, Bank of America, Morgan Stanley, Fiserv, Yahoo!, and iGATE.

TECHNICAL SKILLS

- *AI & GenAI Platforms*: Agentic AI & multi-agent orchestration (LangGraph, Loop Engineering), Model Context Protocol (MCP), LLM observability & tracing (LangSmith), eval harnesses, tool use/function calling, enterprise AI code migration, Claude API, GitHub Copilot.
- *Data & Storage Infrastructure*: Distributed systems, caching architectures, database throughput optimization, multi-AZ resiliency & failover, data migration, schema compatibility, Kafka, payload & logging optimization, REST APIs, GraphQL.
- *Performance & Capacity Engineering*: Continuous profiling (Grafana Pyroscope, Google Cloud Profiler), CPU pinning, memory optimization, cross-language compression (Zstandard, Snappy/LZ4, zlib-rs), throughput & RTT latency analysis, peak stress testing, benchmarking.
- *Site Reliability Engineering*: Distributed systems governance, SLO & error budgeting, rate limiting, traffic shaping, service mesh (Istio), Kubernetes, Knative, observability (Grafana, Golden Signals), deployment readiness gates & rollback planning, CI/CD.
- *Test Engineering & Validation*: TDD, integration & mock-based testing, service virtualization (Tanuki), GenAI-powered E2E test generation, parity validation.
- *Technology Leadership & Governance*: Enterprise platform strategy, multi-tier systems modernization, cross-org functional program leadership (190+ Tier-1/Tier-2 services), engineering standards & OSS governance, strategic roadmapping, OKR/KPI tracking.
- *Programming Languages*: Java, Python, Go, Rust, SQL, Unix scripting.
- *Tools & Platforms*: Grafana, OpenRewrite, Dark Canary (Production Shadowing), JIRA, Confluence.

EXPERIENCE

Walmart: Technical Program Manager – Leadership (Sunnyvale)

Jul 2021 – Current

- Defined and presented Walmart's Java Modernization strategy to VP/SVP/EVP/Chief Architect, establishing an MCP-powered modernization framework that reduced JDK upgrade timelines from 1–2 sprints to 0.25–4 hours across 32K+ production deployments.
- Established Virtual Threads as an org-wide modernization standard, delivering 33% infrastructure optimization and 14.2x throughput improvement for high-scale I/O workloads.
- Transformed enterprise codebase assessment by establishing an MCP-powered code discovery standard, eliminating ~50% redundant code and uncovering ~60% dead code across production repositories at scale.
- Established Walmart's enterprise-standard Node/Java-to-Rust migration framework, standardizing Dark Canary for production traffic shadowing and defining rollout readiness gates and validation criteria across active service migrations and scaling Walmart's first production Rust GraphQL subgraph to 100% US tenant traffic.
- Architected Walmart's cross-language continuous profiling and runtime strategy using Grafana Pyroscope—enabling 40% higher throughput and 87.5% lower memory consumption across Rust services—while establishing standards for CPU pinning, call pattern analysis, and polyglot wire compression (Zstandard for Java, Snappy/LZ4 for Go, and zlib-rs for Rust).
- Expanded Walmart's distributed systems footprint by publishing Go MegaCache, a caching platform delivering 750 TPS/pod, 7–9ms cache-hit latency, and 2x Cassandra throughput while driving enterprise Go adoption across strategic initiatives.
- Established Tanuki as Walmart's service virtualization standard, driving integration-test governance, mock infrastructure, and adoption across 503 Commerce Experience Platform repositories.
- Directed performance engineering governance across 195 holiday-critical real-time services – forecasting peak demand and enforcing strict rate limiting, round-trip time SLAs, payload/logging optimization, and error budgets to remediate 2,793 non-compliant endpoints ahead of peak retail traffic.
- Defined Walmart's Commerce Platform Multi-AZ capacity and resiliency strategy, driving adoption across 15K+ VMs and 158K+ cores to institutionalize sub-5-minute automated failover as the baseline production standard.

Microsoft: Technical Program Manager (Redmond)

Jul 2019 – Jul 2021

- Rearchitected a budgeting and cost management platform, improving performance, data reliability, and forecasting capabilities.
- Integrated Power BI and enterprise data sources to deliver real-time financial, procurement, and Azure cost insights.
- Boosted platform adoption by 200% through advanced cost analytics and embedded Azure pricing capabilities.

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Royal Bank of Canada: Sr. Project Manager (New York)

May 2018 – Jun 2019

- Led BI strategy and delivery, transforming business requirements into scalable analytics solutions.
- Enhanced reporting performance through data modeling and optimization.
- Drove Agile execution, aligning stakeholders and accelerating delivery.

Bank of America: Project Manager - Lead (New York)

Oct 2017 – Apr 2018

- Led enterprise application rationalization, decommissioning 20% of targeted apps to deliver \$19M in savings and establish a roadmap to retire 560 applications with \$512M projected cost reduction.

Morgan Stanley: Project Manager - Lead (New York)

Mar 2013 – Jun 2017

- Led firm-wide Fieldglass/SAP deployment, streamlining workforce management across business units.
- Delivered \$15M in savings through resource optimization, systems integration, and contract negotiation.

Fiserv: Project Manager Lead (New York)

Jan 2012 – Feb 2013

- Architected and delivered an integrated lending platform for banks, connecting Fiserv account processing systems with client environments while managing end-to-end change implementation.

Bank of America: Sr. Project Manager (New York)

Mar 2010 – Dec 2011

- Built trade reconciliation reporting and analytics for FED, providing visibility into valuation, counterparty, and notional trade coverage.
- Led end-to-end delivery of a trade reconciliation platform, driving requirements, planning, risk mitigation, and cross-functional execution.

Yahoo!, Financial Product Manager – Revenue (New York)

Sep 2009 – Feb 2010

- Drove revenue growth insights for top advertisers through competitive and account performance analysis.
- Optimized advertiser revenue forecasts by analyzing CPC, CPM, and CPV models partnering with sales.

iGATE, Sr. Software Engineer (Bangalore, India)

Sep 2004 – Aug 2007

- Engineered enterprise data warehouse solutions by integrating DB2 and Unix-based source systems, enabling scalable analytics and business reporting.
- Automated end-to-end ETL pipelines and data workflows, improving data quality, operational efficiency, and reporting reliability across the organization.

EDUCATION

MBA, University of Connecticut CT

2009

B. E. Computer Science, Oriental Institute of Science & Technology, INDIA

2003